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Project Work Synopsis

On

"AI-Based Intelligent Surveillance System"

SUBMITTED IN PARTIAL FULFILLMENT FOR THE AWARD OF DEGREE OF

BACHELOR OF ENGINEERING

IN

Computer Science & Engineering

SUBMITTED BY

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SYNOPSIS APPROVAL STATEMENT

The project synopsis titled "**AI-Based Intelligent Surveillance System**" has been reviewed and approved by the undersigned project guide and Head of the Department. The proposed work is found to be relevant, technically feasible, and aligned with the academic objectives and curriculum requirements of the program. The problem statement, objectives, and proposed methodology demonstrate sufficient scope for learning and practical implementation. Hence, the synopsis is accepted, and the students are permitted to proceed with the project work under the guidance of the allotted project guide.

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Abstract

Suspicious activities pose significant threats to public safety, financial systems, and organizational security, making accurate and timely detection essential for effective risk management and crime prevention. Traditional methods of monitoring and identifying suspicious behavior often rely on manual surveillance, rule-based systems, and human judgment, which can be time-consuming, inconsistent, and prone to errors due to the increasing volume and complexity of data. To address these challenges, this project proposes an **AI-Based Intelligent Surveillance System** that leverages artificial intelligence and data-driven techniques to provide accurate, real-time, and intelligent threat detection.

The proposed system integrates machine learning and artificial intelligence algorithms with diverse data sources such as video surveillance feeds, transaction records, user activity logs, network traffic data, and historical incident reports. The AI system acts as an intelligent monitoring platform that not only detects unusual or suspicious patterns but also provides alerts and insights to security personnel for timely intervention. By analyzing large volumes of structured and unstructured data, the system identifies hidden patterns, anomalies, and behavioral deviations that may indicate potential security risks.

Key components of the system include data collection and preprocessing, feature extraction, model training, anomaly detection, and a user-friendly monitoring interface. Machine learning models such as classification algorithms, anomaly detection techniques, decision trees, or deep learning networks are employed to enhance detection accuracy and reduce false positives. The AI system further improves usability by allowing administrators to receive automated alerts, risk assessments, and detailed activity reports, thereby minimizing reliance on continuous manual supervision.

The proposed solution aims to strengthen decision-making in areas such as fraud prevention, cyber threat detection, physical security monitoring, and risk assessment. It also helps mitigate risks associated with criminal activities, financial fraud, data breaches, and unauthorized access.

In conclusion, the AI-Based Intelligent Surveillance System represents a smart and scalable approach to modern security challenges. By combining artificial intelligence with advanced data analytics, the system enhances proactive threat detection, improves response efficiency, and contributes to the broader goal of ensuring safety, security, and trust in digital and physical environments.

Introduction

With the rapid growth of urbanization and the increasing deployment of surveillance cameras in public and private environments, large volumes of video data are generated continuously. Monitoring these video feeds manually is inefficient, time-consuming, and prone to human error, as security personnel may miss important incidents due to fatigue, distraction, or simultaneous monitoring of multiple screens. Therefore, there is a growing demand for intelligent automated surveillance systems capable of detecting suspicious activities in real time and providing timely alerts.

Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision technologies have significantly transformed the field of surveillance by enabling automated analysis of visual data. AI-based systems can process video streams, identify objects, track human movements, and recognize abnormal behaviour patterns with high accuracy. These advancements support the development of smart surveillance systems that reduce dependence on human monitoring while improving response time to potential threats. Recent research demonstrates that deep learning models such as convolutional neural networks (CNNs) and object detection frameworks can effectively detect suspicious activities such as loitering, violence, intrusion, and abandoned objects.

In this context, the AI Suspicious Activity Detection System is proposed as an intelligent solution for automated security monitoring. The system integrates computer vision algorithms with real-time video processing to detect suspicious behaviour patterns from CCTV feeds. It also includes a web-based dashboard for visualization, alert generation, and event logging to support decision-making by security personnel. Additionally, the system aims to detect medical emergency situations such as sudden falls or collapse of individuals, enabling faster response and assistance in crowded environments.

The primary objective of this project is to design and develop a scalable, efficient, and userfriendly AI-based surveillance system capable of improving security monitoring and reducing manual effort. By providing automated detection and real-time alerts, the proposed system contributes toward safer environments in public spaces, institutions, workplaces, and smart city infrastructures.

Literature Survey

1. Suspicious Activity Detection in Surveillance Systems

Suspicious activity detection has become a critical component of intelligent surveillance systems, particularly with the increasing deployment of CCTV cameras in public places such as airports, railway stations, shopping malls, campuses, and urban infrastructures. The primary objective of such systems is to automatically identify abnormal or unusual behaviour patterns that may indicate potential threats, criminal activities, or unsafe situations. Traditional surveillance approaches rely heavily on human operators to monitor video feeds continuously, which often results in missed incidents due to fatigue, limited attention span, and cognitive overload. This limitation has created a strong demand for automated surveillance solutions that can assist or replace manual monitoring.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and Big Data analytics have enabled the development of intelligent surveillance systems capable of analysing large volumes of video data efficiently. AI-based systems can learn behavioural patterns from historical data and detect anomalies by comparing real-time activities with learned normal patterns. These techniques are particularly useful in identifying suspicious behaviours such as loitering, unauthorized access, intrusion, abandoned objects, and violent activities. Research studies indicate that automated surveillance systems significantly improve monitoring efficiency, reduce response time, and enhance security management compared to traditional methods.

Furthermore, modern surveillance research is increasingly focusing on integrating multimodal data sources, including video, sensor data, and contextual information such as time and location, to improve detection accuracy. Intelligent surveillance systems are also being incorporated into smart city infrastructures to enhance public safety and resource management. These developments demonstrate the growing importance of AI-powered suspicious activity detection systems in modern security applications.

2. Machine Learning Approaches for Activity Classification

Machine learning techniques have played a significant role in the development of suspicious activity detection systems by enabling automated classification of behavioural patterns. Traditional machine learning algorithms such as Support Vector Machines (SVM), Decision

Trees, Artificial Neural Networks (ANN), K-Nearest Neighbors (KNN), and Naïve Bayes classifiers have been widely used to detect anomalies and classify suspicious activities based on extracted features. These models analyze patterns from historical datasets and predict whether a particular activity is normal or abnormal.

Vorapatratorn (2024) proposed an AI-based suspicious activity monitoring system that integrates event-related data from individuals, vehicles, and objects using multiple machine learning algorithms. The study compared various classification methods and reported that Decision Tree algorithms achieved the highest accuracy of approximately 98.86%, demonstrating their effectiveness for structured suspicious activity classification tasks. Such results highlight the potential of machine learning algorithms in surveillance applications, especially when sufficient labelled data is available.

However, traditional machine learning approaches have certain limitations. Most models rely on handcrafted features and structured datasets rather than raw video streams, which restricts their performance in real-time dynamic environments. Additionally, these models often struggle to capture complex temporal relationships present in video data, such as movement patterns over time. As a result, researchers are increasingly shifting toward deep learning approaches that can automatically extract meaningful features from visual data and provide improved performance in real-time surveillance scenarios.

3. Deep Learning and Computer Vision Techniques

Deep learning and computer vision technologies have significantly advanced the capabilities of intelligent surveillance systems by enabling automated object detection, tracking, and behaviour recognition directly from video data. Convolutional Neural Networks (CNNs) have proven particularly effective in extracting spatial features from images and video frames, allowing systems to detect objects such as humans, vehicles, and suspicious items with high accuracy. Among the most widely used object detection frameworks is YOLO (You Only Look Once), introduced by Redmon et al., which provides real-time detection with high speed and precision, making it suitable for surveillance applications.

In addition to object detection, tracking algorithms such as DeepSORT (Deep Simple Online and Realtime Tracking) have been developed to maintain identity information of detected objects across video frames. This capability is essential for monitoring individual behaviour

over time, especially in crowded environments where multiple people are present. Combining object detection with tracking allows systems to analyze movement trajectories, interaction patterns, and behavioural changes that may indicate suspicious activities.

Recent research has also explored hybrid deep learning architectures such as CNN-LSTM networks, which combine spatial feature extraction with temporal sequence analysis. These models enable systems to understand complex behavioural patterns by analysing motion over time rather than relying solely on individual frames. Such approaches have shown improved accuracy in detecting abnormal activities such as violence, intrusion, and loitering. The integration of deep learning and computer vision technologies continues to enhance the performance and reliability of intelligent surveillance systems.

4. Fall Detection and Medical Emergency Monitoring

Beyond security monitoring, intelligent surveillance systems are increasingly being used to detect medical emergencies such as falls, sudden collapse, or abnormal posture changes. In crowded public environments, individuals may experience health-related incidents such as fainting, heart attacks, or injuries that may go unnoticed due to limited human supervision. Automated fall detection systems can play a crucial role in identifying such incidents and providing timely alerts for emergency assistance.

Recent research in computer vision has focused on pose estimation techniques, which analyze human body keypoints to understand posture and movement patterns. By monitoring changes in body orientation and motion trajectories, AI systems can detect abnormal events such as sudden falls or prolonged inactivity. Deep learning models trained on human activity datasets have demonstrated promising results in recognizing fall events with high accuracy.

Integrating fall detection with suspicious activity monitoring enhances the overall effectiveness of surveillance systems by addressing both security and health-related concerns. This dual functionality increases the practical applicability of intelligent monitoring systems in environments such as hospitals, elderly care centres, campuses, offices, and public transportation hubs. As AI technologies continue to evolve, fall detection systems are expected to become more accurate and reliable, contributing to improved public safety and emergency response systems.

5. Web-Based Monitoring and Visualization Systems

Modern intelligent surveillance systems often incorporate web-based dashboards and visualization platforms to support monitoring, analysis, and decision-making processes. These dashboards provide real-time visualization of detected events, statistical summaries, and alerts, enabling security personnel to respond quickly to potential threats. Research studies indicate that integrating visualization tools with AI-based detection systems significantly improves usability and operational efficiency.

Web-based monitoring platforms typically include features such as live video streaming, event logs, screenshots of detected incidents, and graphical representations of activity patterns. Geographic mapping and time-based analytics further enhance situational awareness by providing contextual information about detected events. Big data technologies such as MySQL, MongoDB, and Apache Hive have been explored for storing and managing large volumes of surveillance data, with scalable database solutions offering better performance for large-scale deployments.

The proposed project extends existing research by integrating real-time computer vision techniques, suspicious behaviour detection, fall detection capabilities, and a web-based visualization interface into a unified intelligent surveillance platform. Additionally, incorporating automated alert notifications, cloud storage support, and scalable architecture improves system reliability and accessibility. Such integrated systems have significant potential for deployment in smart cities, educational institutions, corporate environments, and transportation infrastructures, contributing toward safer and more efficient surveillance management.

6. Explainability and Decision Support

As AI-based surveillance systems are increasingly deployed in public safety, smart city infrastructure, and institutional environments, the interpretability of machine learning models has become an important research focus. Security personnel and decision-makers must understand why a system has classified a behaviour as suspicious in order to trust the technology and take appropriate action. Recent studies have explored explainable AI (XAI) techniques such as feature importance analysis, attention visualization, and model confidence

scoring to improve transparency in activity recognition systems. These methods help identify which behavioural patterns, motion trajectories, or contextual features contributed to the detection decision.

Researchers have also emphasized the importance of integrating AI predictions with decision-support mechanisms such as automated alerts, risk scoring, and event summaries. Combining AI detection with contextual information like time, location, crowd density, and historical incident patterns can enhance situational awareness and improve response planning. Such intelligent decision-support systems enable security personnel to prioritize threats, allocate resources effectively, and respond more quickly to critical incidents.

The proposed AI Suspicious Activity Detection System addresses these challenges by incorporating interpretable detection outputs, visual evidence such as bounding boxes and activity timelines, and a user-friendly monitoring interface. By integrating real-time analytics with explainable insights and automated alerts, the system aims to improve user trust, operational efficiency, and decision-making effectiveness. Drawing from recent advancements in explainable AI and intelligent surveillance research, the project seeks to deliver a reliable, transparent, and practical solution for modern security monitoring environments.

Existing System

In the current surveillance environment, monitoring of CCTV cameras is primarily performed manually by security personnel. Operators are required to continuously observe multiple video feeds displayed across several screens in order to identify suspicious behaviour or potential security threats. While this approach has been widely used for many years, it suffers from several inherent limitations such as human fatigue, limited attention span, and decreased concentration over long monitoring periods. As the number of cameras increases in modern infrastructures such as campuses, transportation hubs, and smart cities, manual monitoring becomes increasingly impractical and inefficient. Security personnel may miss critical incidents due to cognitive overload, resulting in delayed responses and compromised safety.

Traditional surveillance systems mainly function as passive recording tools that store video footage for later review rather than providing proactive monitoring or automated analysis. In many cases, suspicious activities are identified only after an incident has occurred by reviewing recorded footage, which reduces the effectiveness of preventive security measures. Although some systems incorporate basic motion detection algorithms, these methods are limited in capability and often generate false alarms due to environmental factors such as lighting variations, shadows, weather changes, or background movement. Such false alerts can reduce system reliability and increase the workload of security staff.

Another major limitation of existing surveillance systems is the absence of intelligent behaviour analysis and real-time decision support. Most systems are not capable of understanding complex human activities such as loitering, abnormal movement patterns, intrusion into restricted areas, or violent actions. Additionally, current surveillance infrastructures typically lack automated alert generation, event logging, and reporting mechanisms that could assist security personnel in responding quickly to potential threats. The absence of advanced analytics also prevents the extraction of meaningful insights from surveillance data, limiting its usefulness for long-term security planning and management.

Furthermore, existing systems rarely incorporate detection of medical emergencies such as falls, sudden collapse, or abnormal posture changes, which are critical in crowded environments like public transport stations, offices, and educational institutions. Without automated detection, such incidents may go unnoticed for extended periods, delaying emergency assistance and increasing risk to individuals.

Overall, existing surveillance systems are largely reactive, manual, and inefficient for largescale monitoring environments. They lack automation, intelligent analysis, real-time alerts, and decision-support capabilities required for modern security needs. These limitations highlight the necessity of developing an AI-based suspicious activity detection system that can automatically analyze video streams, accurately detect abnormal behaviours, and provide timely alerts and actionable insights for improved safety and operational efficiency.

Problem Statement

- Effective surveillance is critical for maintaining safety and security in public and private spaces such as smart cities, campuses, transportation systems, and commercial areas.
- Existing surveillance systems depend heavily on manual monitoring, which is inefficient and prone to human errors caused by fatigue, distraction, and the difficulty of observing multiple video feeds simultaneously.
- The rapid increase in the number of surveillance cameras has made manual monitoring even more challenging and less reliable.
- Many current AI-based surveillance solutions focus mainly on static datasets instead of real-time video analysis, limiting their effectiveness in dynamic real-world environments.
- Existing systems often struggle to continuously monitor live video streams, understand behavioural patterns over time, and respond immediately to suspicious activities.
- High false alarm rates, lack of adaptability to different environments, and poor integration with alert and reporting systems further reduce the effectiveness of current solutions.
- The absence of intelligent automation delays response time to critical incidents and increases dependency on human supervision.
- There is a strong need for an automated intelligent surveillance system capable of accurately detecting suspicious behaviour in real time and generating alerts without constant human monitoring.
- Such a system should provide real-time notifications, event logs, and visual evidence to support quick decision-making by security personnel.
- The proposed project aims to address these challenges by developing an AI-powered suspicious activity detection system using computer vision and machine learning to improve monitoring efficiency, reduce workload, enhance response time, and create safer environments.

Research Gaps

Despite significant advancements in AI Suspicious Activity Detection System, several research gaps still remain that limit the practical adoption and effectiveness of existing systems:

Despite advancements in AI-based surveillance systems, several research gaps still exist:

a. Limited Real-Time Implementation

Many systems focus on offline analysis rather than real-time video processing.

b. Insufficient Behaviour Understanding

Existing models often detect objects but fail to accurately interpret complex human behaviour patterns.

c. High Computational Requirements

Deep learning models require significant computational resources, limiting deployment on low-cost hardware.

d. Lack of Integrated Emergency Detection

Few systems combine suspicious activity detection with medical emergency detection such as falls.

e. False Alarm Issues

Environmental variations often lead to incorrect detection results.

f. Limited User-Friendly Interfaces

Many research systems lack practical dashboards for monitoring and decision support.

Addressing these gaps is essential for developing an effective AI-based surveillance system.

Motivation

Public safety and security have become major concerns in modern society due to the increasing number of surveillance cameras deployed in public and private spaces such as campuses, transportation hubs, offices, and smart city environments. Security personnel are often required to monitor multiple video feeds continuously, which can lead to fatigue, reduced attention, and delayed detection of suspicious activities. Critical incidents such as unauthorized access, violent behaviour, or medical emergencies may go unnoticed, resulting in potential risks to individuals and infrastructure. The motivation behind this project arises from the need to develop intelligent automated systems that can support security monitoring by providing accurate, timely, and reliable detection of abnormal activities.

Recent advancements in Artificial Intelligence, Machine Learning, and Computer Vision technologies provide an opportunity to transform traditional surveillance systems into intelligent monitoring solutions. By leveraging deep learning techniques, it is possible to analyze large volumes of video data, identify behavioural patterns, and detect suspicious activities more efficiently than manual observation. Developing an AI Suspicious Activity Detection System is motivated by the objective of bridging the gap between advanced AI technologies and practical security applications by creating a system that is efficient, scalable, and easy to use in real-world environments.

Another key motivation is the lack of integrated surveillance systems that not only detect suspicious activities but also provide meaningful insights and automated alerts for decisionmaking. An intelligent AI-based system can assist security personnel by highlighting potential threats, generating alerts, and maintaining event logs, thereby reducing workload and improving response time. Additionally, incorporating fall detection and medical emergency monitoring increases the system’s usefulness by addressing both security and safety concerns. Ultimately, this project is motivated by the goal of enhancing public safety, improving surveillance efficiency, and promoting the adoption of intelligent technologies in modern security infrastructures.

Objectives of the Project

The primary objectives of the AI Suspicious Activity Detection System are as follows:

1. To develop an AI-based intelligent surveillance system for real-time activity monitoring: The main objective of the project is to design and implement an automated surveillance system capable of monitoring live video streams and detecting suspicious or abnormal activities in real time. The system aims to reduce dependency on manual monitoring while improving detection accuracy, efficiency, and response time in dynamic environments.

2. To implement object detection, tracking, and behavioural analysis using deep learning techniques: The project aims to utilize advanced deep learning models such as YOLO for object detection and tracking algorithms like DeepSORT to continuously monitor individuals across video frames. It also focuses on analyzing behavioural patterns, motion trajectories, and temporal activities to accurately identify abnormal situations such as intrusion, loitering, violent movements, abandoned objects, or unusual behaviour.

3. To generate automated alerts and detect emergency situations including fall detection: Another objective is to develop a real-time alert mechanism that provides notifications, screenshots, timestamps, and event details whenever suspicious activities or medical emergencies such as falls or sudden collapse are detected. This feature enhances safety, enables quick response, and improves the overall effectiveness of monitoring systems.

4. To develop a web-based dashboard for monitoring, visualization, and system management: The project also aims to create a user-friendly web interface that displays live video monitoring results, detected events, activity logs, and statistical summaries. The dashboard will support efficient monitoring, reporting, and decision-making while providing administrators with centralized control over the surveillance system.

Tools (Software and Hardware Requirements)

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1. Software Requirements

i. Operating System : Windows 10 / Windows 11 or Ubuntu Linux

Provides a stable environment for development, training, and deployment of AI-based computer vision models.

ii. Programming Language : Python

Chosen for its simplicity and extensive support for machine learning, deep learning, and image processing libraries.

iii. Development Environment / IDE : Jupyter Notebook / Visual Studio Code / PyCharm

Used for coding, model development, debugging, testing, and visualization of results.

iv. Machine Learning & AI Libraries

- NumPy, Pandas – data processing and numerical computations
- OpenCV – image and video processing
- TensorFlow / PyTorch – deep learning model development
- YOLO Framework – object detection
- Matplotlib / Seaborn – data visualization and performance analysis

v. Database Management System : MySQL / SQLite / MongoDB

Used for storing detected event logs, timestamps, screenshots, and monitoring data.

vi. Web/Application Framework : Flask / Django / Streamlit

Used to develop a user-friendly dashboard interface for monitoring suspicious activities and alerts.

vii. APIs and External Data Sources : Public video datasets (e.g., surveillance datasets), Camera input APIs

Used for training, testing, and integrating real-time video feeds into the system.

2. Hardware Requirements

Development System

i. Processor: Intel i5 or higher

ii. RAM: Minimum 8 GB (16 GB recommended) iii. Storage: Minimum 256 GB SSD

Used for model training, testing, and application development.

iv. Camera Devices

▪ CCTV Camera / Webcam

Used to capture live video streams for suspicious activity detection.

v. GPU (Optional but Recommended)

NVIDIA GPU or equivalent for faster deep learning model training and real-time processing.

vi. Embedded / Edge Device (Optional)

NVIDIA Jetson Nano / Raspberry Pi

Interfaces with cameras and enables deployment of the system in real-world environments.

References

[1] J. Redmon, S. Divvala, R. Girshick and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2016.O. Adedeji, B. Ghimire, R. Karn, E. Wieber, A. Abdalla and W. Guo, "Cotton yield prediction using satellite remote sensing and machine learning," Proc. 2023 IEEE Int. Geosci. Remote Sens. Symp. (IGARSS), Pasadena, CA, USA, 2023.

[2] N. Wojke, A. Bewley and D. Paulus, "Simple Online and Realtime Tracking with a Deep Association Metric," Proc. IEEE Int. Conf. Image Processing (ICIP), 2017.

[3] S. Vorapatratorn, "Enhancing Monitoring of Suspicious Activities with AI-Based and Big Data Fusion," PeerJ Computer Science, 2024.

[4] W. Liu et al., "SSD: Single Shot MultiBox Detector," Proc. European Conf. Computer Vision (ECCV), 2016.

[5] A. Krizhevsky, I. Sutskever and G. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," Advances in Neural Information Processing Systems (NIPS), 2012.Proc. 2023 Int. Conf. Ambient Intell., Knowl. Informatics Ind. Electron. (AIKIIE), Warangal, India, pp. 1–5, 2023.

[6] M. Everingham et al., "The Pascal Visual Object Classes (VOC) Challenge," Int. Journal of Computer Vision, 2015.

[7] I. Goodfellow, Y. Bengio and A. Courville, "Deep Learning," MIT Press, 2016.

[8] Z. Cao, T. Simon, S. Wei and Y. Sheikh, "Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields," Proc. CVPR, 2017.

[9] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," arXiv preprint, 2014.

[10] C. Luo, X. Zhao and W. Huang, "Abnormal Event Detection in Surveillance Videos Using Deep Learning," IEEE Access, 2020.

[11] R. Mehran, A. Oyama and M. Shah, "Abnormal Crowd Behavior Detection Using Social Force Model," Proc. CVPR, 2009.

[12] H. Hassner, Y. Itcher and O. Kliper-Gross, "Violent Flows: Real-Time Detection of Violent Crowd Behavior," Proc. CVPR Workshops, 2012.

[13] S. Sultani, C. Chen and M. Shah, "Real-World Anomaly Detection in Surveillance Videos," Proc. CVPR, 2018.

[14] T. Bouwmans, "Traditional and Recent Approaches in Background Modeling for Foreground Detection," Computer Science Review, 2014.

[15] OpenCV Documentation, "Open Source Computer Vision Library," Available: https://opencv.org

[16] A. Doshi and M. Yilmaz, "Fall Detection Using Computer Vision: A Review," Expert Systems with Applications, 2021.